In [1]:
```python
# step 1 import libraries

import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:
```python
# step 2 load dataset

data = pd.read_csv("data/book_data.csv")
data.columns = data.columns.str.strip()
data
```

Out[2]:

	Book_ID	Title	Author	Pages	Price	Rating	Year
0	1	Python Basics	John	250	350	4.5	2020
1	2	ML Guide	Smith	400	550	4.8	2021
2	3	Data Science	Alex	320	450	4.3	2019
3	4	Neural Network	Smith	415	460	4.5	2020
4	5	Deep Learning	Marsland	350	390	4.6	2022
5	6	NLP	Michele	255	480	3.9	2023

In [3]:
```python
# step 3 display dataset info

data.info()
data.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 7 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   Book_ID  6 non-null      int64
 1   Title    6 non-null      object
 2   Author   6 non-null      object
 3   Pages    6 non-null      int64
 4   Price    6 non-null      int64
 5   Rating   6 non-null      float64
 6   Year     6 non-null      int64
dtypes: float64(1), int64(4), object(2)
memory usage: 468.0+ bytes
```

Out[3]:

	Book_ID	Pages	Price	Rating	Year
count	6.000000	6.000000	6.000000	6.000000	6.000000
mean	3.500000	331.666667	446.666667	4.433333	2020.833333
std	1.870829	70.190218	70.047603	0.307679	1.471960
min	1.000000	250.000000	350.000000	3.900000	2019.000000
25%	2.250000	271.250000	405.000000	4.350000	2020.000000
50%	3.500000	335.000000	455.000000	4.500000	2020.500000
75%	4.750000	387.500000	475.000000	4.575000	2021.750000
max	6.000000	415.000000	550.000000	4.800000	2023.000000

In [4]:

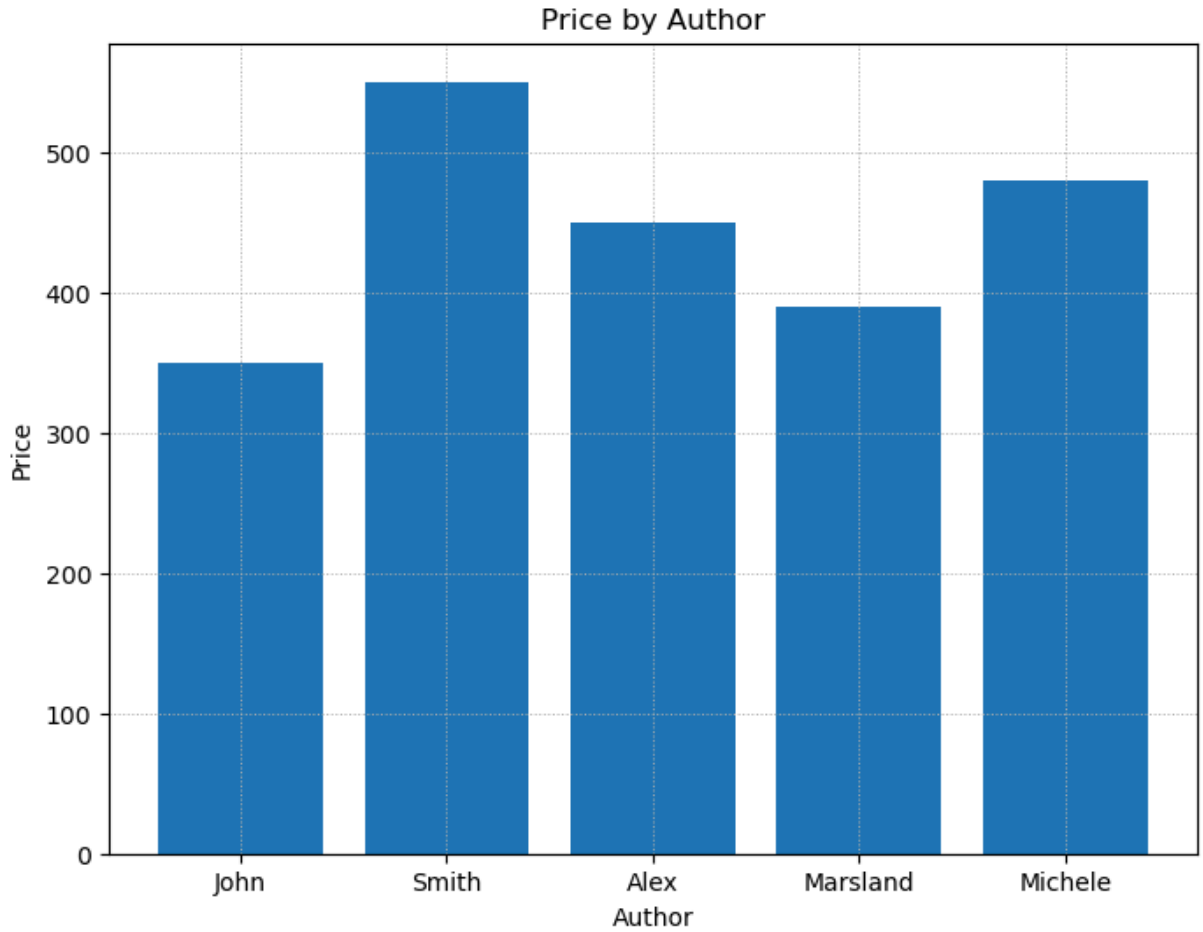
```
# step 4 plot bar graph

plt.figure(figsize=(8,6))

plt.bar(data['Author'], data['Price'])

plt.xlabel("Author")
plt.ylabel("Price")
plt.title("Price by Author")

plt.grid(True, linestyle=':', linewidth=0.7)
plt.savefig("bar_graph.png", dpi=300, bbox_inches="tight")
plt.show()
```



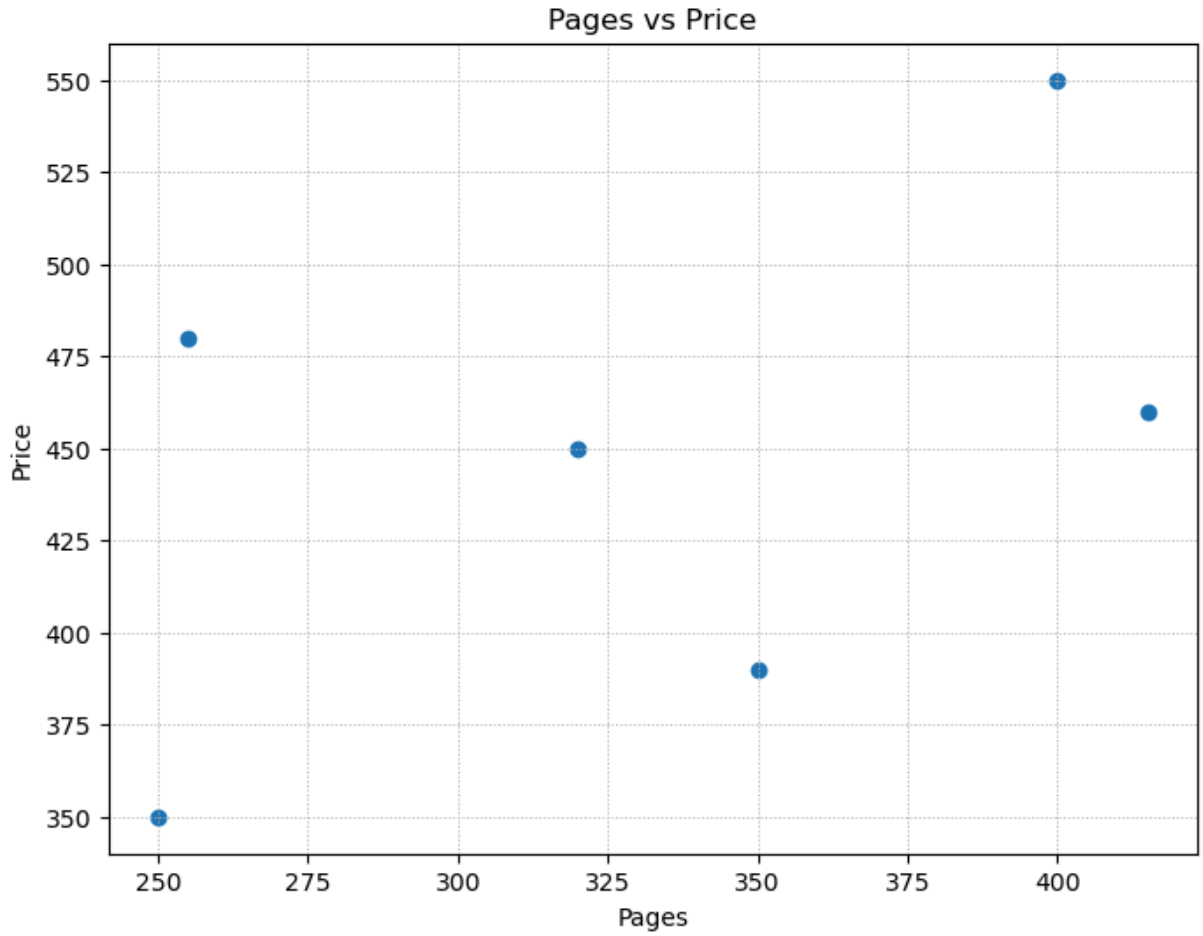
In [5]:
```
# step 5 plot scatter plot

plt.figure(figsize=(8,6))

plt.scatter(data['Pages'], data['Price'])

plt.xlabel("Pages")
plt.ylabel("Price")
plt.title("Pages vs Price")

plt.grid(True, linestyle=':', linewidth=0.7)
plt.savefig("scatter_plot.png", dpi=300, bbox_inches="tight")
plt.show()
```



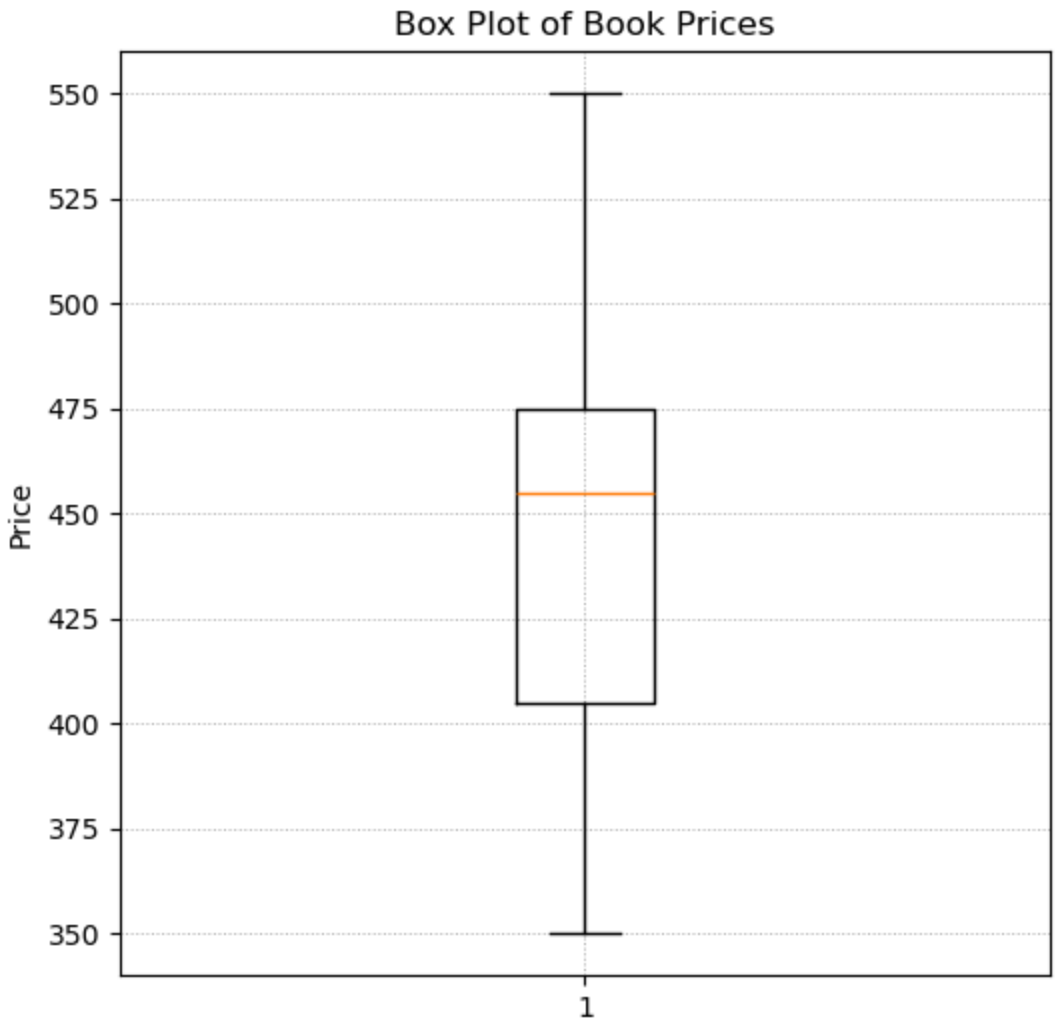
In [6]:
```
# step 6 plot box plot

plt.figure(figsize=(6,6))

plt.boxplot(data['Price'])

plt.ylabel("Price")
plt.title("Box Plot of Book Prices")

plt.grid(True, linestyle=':', linewidth=0.7)
plt.savefig("box_plot.png", dpi=300, bbox_inches="tight")
plt.show()
```



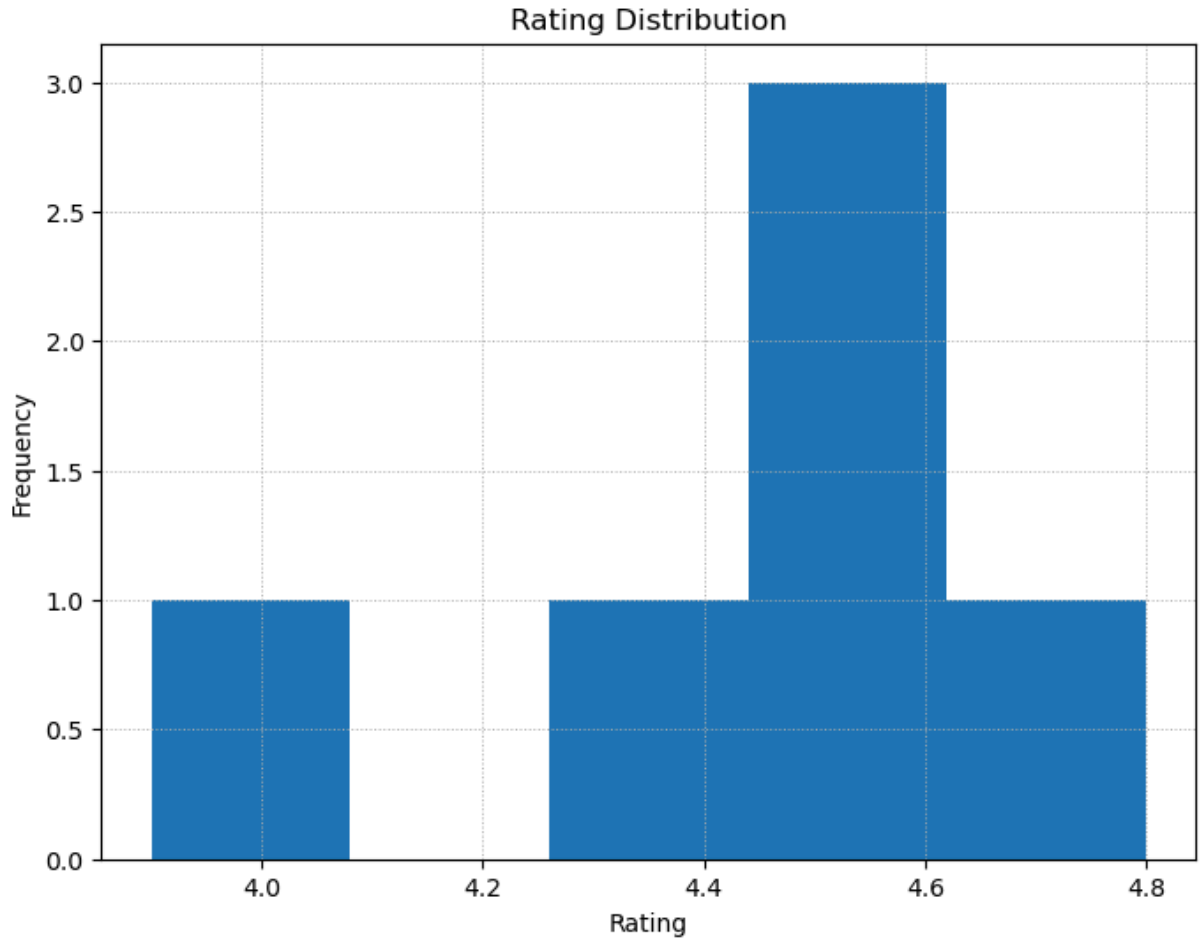
In [12]:
```
# step 7 plot histogram

plt.figure(figsize=(8,6))

plt.hist(data['Rating'], bins=5)

plt.xlabel("Rating")
plt.ylabel("Frequency")
plt.title("Rating Distribution")

plt.grid(True, linestyle=':', linewidth=0.7)
plt.savefig("histogram.png", dpi=300, bbox_inches="tight")
plt.show()
```



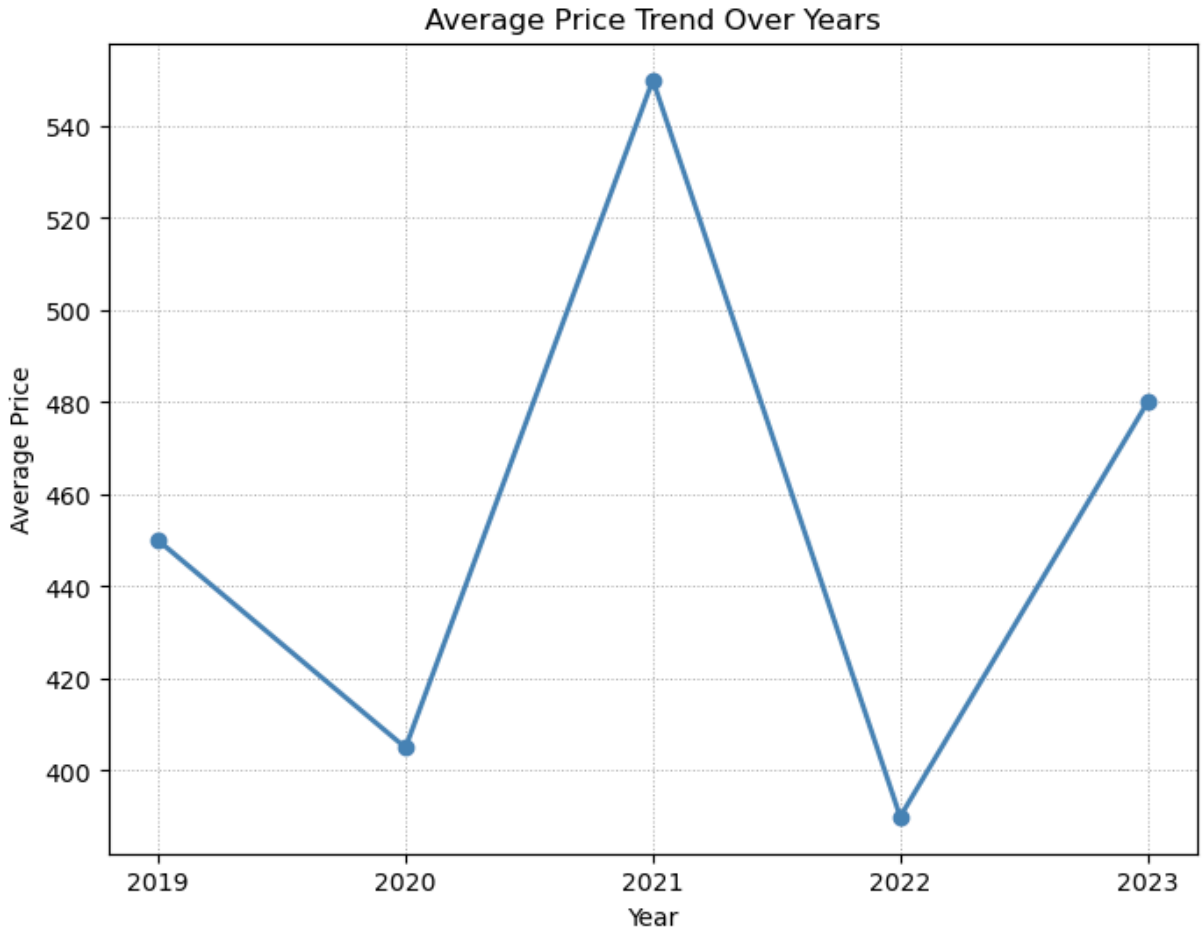
In [10]:
```
# step 8 plot line graph

plt.figure(figsize=(8, 6))

yearly_avg = data.groupby('Year')['Price'].mean().reset_index()

plt.plot(yearly_avg['Year'], yearly_avg['Price'], marker='o', linewidth=2, c

plt.xlabel("Year")
plt.ylabel("Average Price")
plt.title("Average Price Trend Over Years")
plt.xticks(yearly_avg['Year'].astype(int))
plt.grid(True, linestyle=':', linewidth=0.7)
plt.savefig("line_graph.png", dpi=300, bbox_inches="tight")
plt.show()
```



In [13]:
```
# step 9 display analysis table

analysis = pd.DataFrame({
    "Graph Type": ["Bar", "Scatter", "Box", "Histogram", "Line"],
    "Observation": [
        "Author A has highest price",
        "Positive correlation",
        "1 outlier found",
        "Most ratings 4–5",
        "Increasing trend"
    ]
})

analysis
```

Out[13]:

	Graph Type	Observation
0	Bar	Author A has highest price
1	Scatter	Positive correlation
2	Box	1 outlier found
3	Histogram	Most ratings 4–5
4	Line	Increasing trend